

Duke Agent Chatbot

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Data Sourcing and Collection

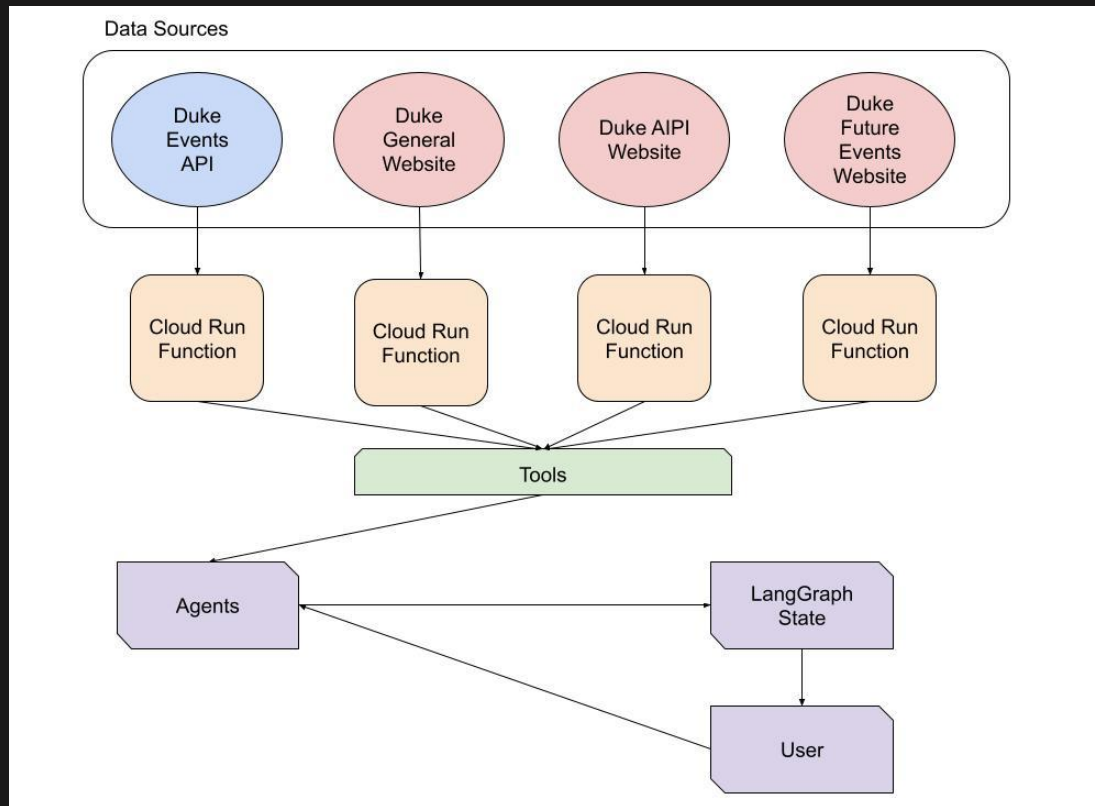
Duke Events API

- Provides streams of data which show a calendar of events at various Duke Campuses.
- The chatbot calls this api when the right tool is called
- <http://urlbuilder.calendar.duke.edu/>

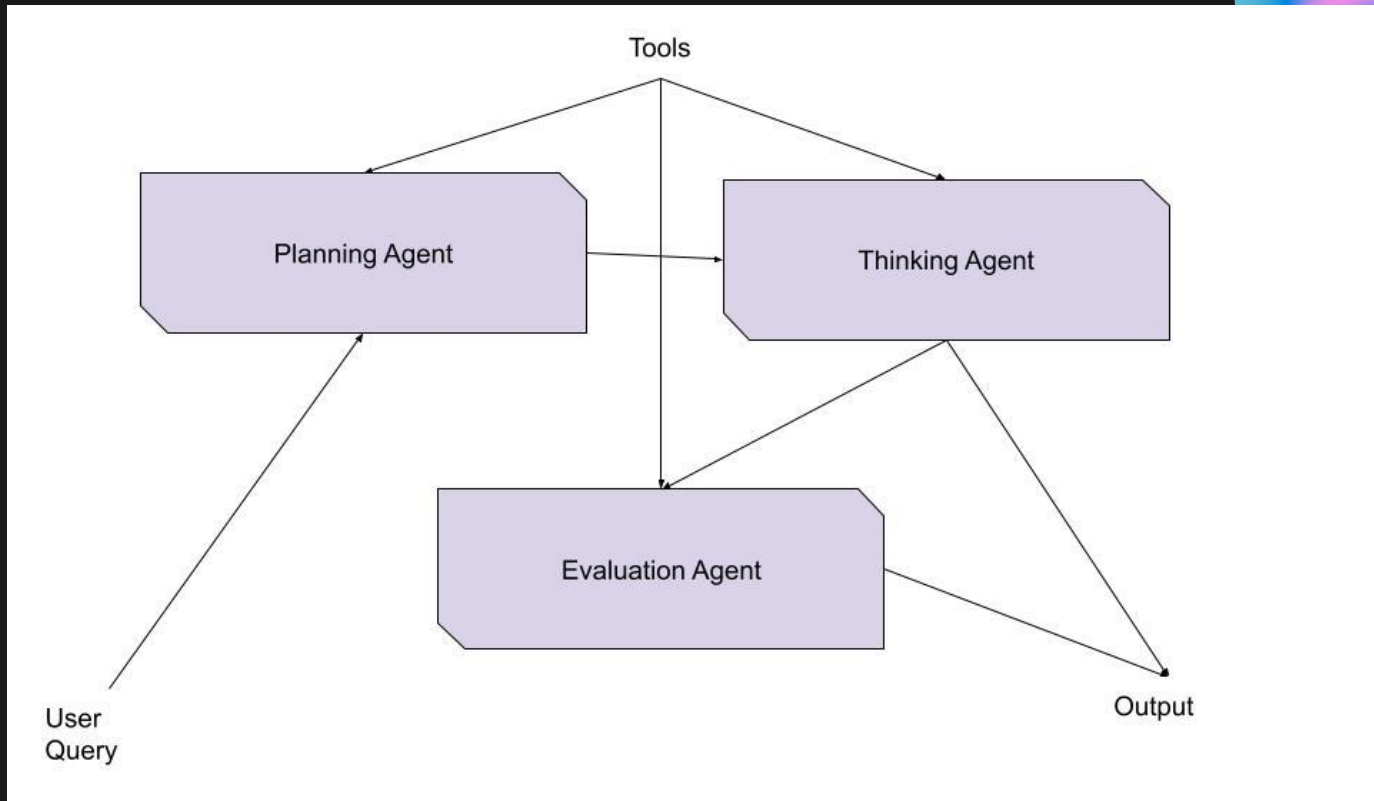
Duke Websites

- We use Google's Programmable Search Engine to allow for our chatbot to make searches
- Websites:
 - masters.pratt.duke.edu/life/students/*
 - masters.pratt.duke.edu/people?s=&department=artificial-intelligence&group=faculty&submit=Filter*
 - masters.pratt.duke.edu/ai/*

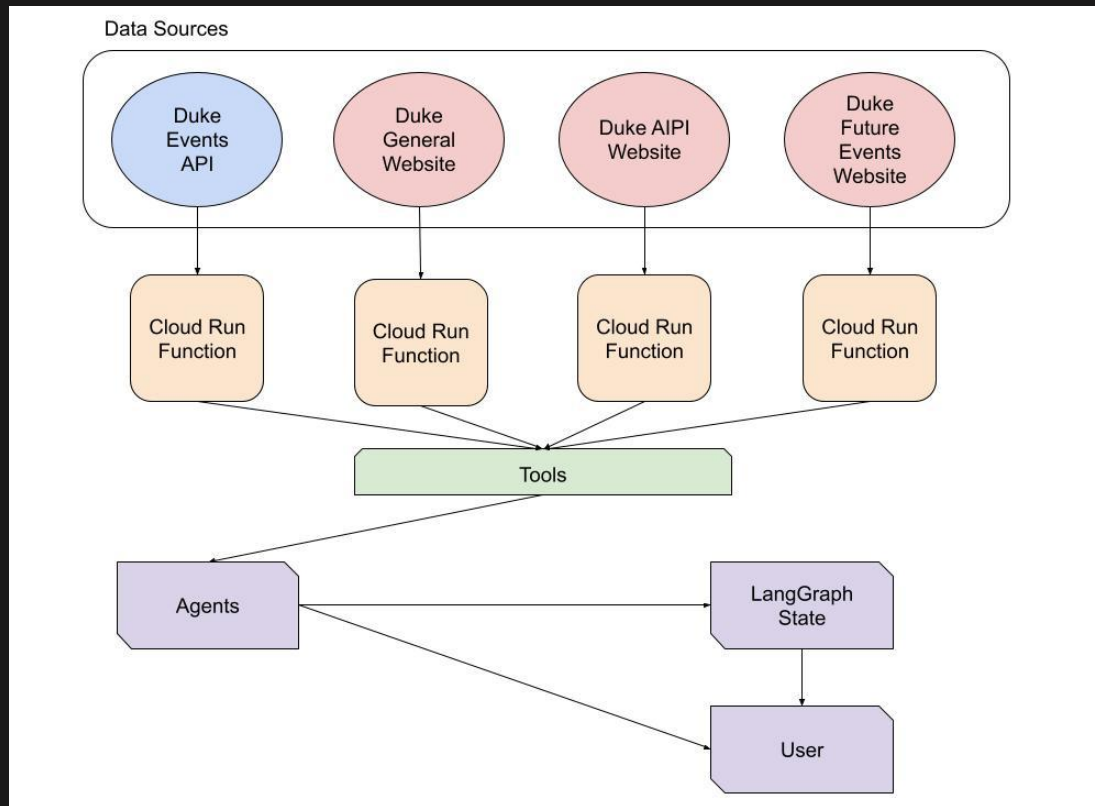
System Architecture



System Architecture



System Architecture



Modeling Decisions

Gemini API

Vertex API

Agentic Framework

- Evaluation: Evaluates the output
- Planning: User takes the user input and passes the tool search and API to the Thinking
- Thinking: Runs the tools

Tools

- DukeAI Tool (uses PSE)
- Duke General (uses PSE)
- Duke Events (uses API call)
- Duke Future Events (uses PSE)

Performance Evaluation: LLM as a Judge

What is the AIPI program and what are the degree requirements?

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The AIPI program at Duke University refers to the Master of Engineering in Artificial Intelligence for Product Innovation (E-EGR-AIPI). This professional master's program within the Pratt School of Engineering focuses on equipping students with the skills to apply AI and machine learning to product development. The program emphasizes an industry-connected curriculum, developed in collaboration with industry partners, ensuring practical relevance and career readiness. While the provided search results don't explicitly list all degree requirements in one place, they offer several crucial links and pieces of information allowing us to piece together a clearer picture.

The program involves a combination of core courses, electives, and project work. Some mentioned courses associated with the AIPI program include AIPI 503: Python Bootcamp, AIPI 505: Mid-Program Residency, and AIPI 504: Introductory Residency. Additionally, students undertake projects that require approval from the AIPI program director. As per one of the search snippets, the core curriculum is geared towards real-world application rather than just theoretical knowledge. This program aims to produce graduates capable of leading in the application of AI and machine learning within a product innovation context. Jon Reifschneider is mentioned as being involved in the program, teaching courses such as AIPI 501, 504, and 505. While his exact title isn't specified in these results, other sources suggest he is significantly involved in the program's direction.

For detailed degree requirements, the best approach would be to consult the official program bulletin linked in the results: <https://prattprofessional.bulletins.duke.edu/programs/E-EGR-AIPI>. This document should contain comprehensive information on credit hours, core courses, electives, project requirements, and any other specific criteria for graduation. The website for Duke Engineering Master's programs (<https://masters.pratt.duke.edu/ai/degree/>) is also listed as a resource for application requirements.

While this response provides a general overview of the AIPI program and points you towards the definitive resources for degree requirements, I encourage you to explore the provided links for the most accurate and up-to-date information. The program pages are the official source and will always have the latest details. Please don't hesitate to ask if you have further questions after reviewing those resources.

Accuracy: 9/10 Relevance: 9/10 Completeness: 8/10

Show thinking process

Ask about Duke University...

Accuracy: 9/10

Relevance: 9/10

Completeness: 8/10

Agent Thinking Process

I saw you were asking about the AIPI program and its degree requirements. I understood this to be a general question about a Duke program looking for information specifically about what it takes to earn the degree.

Because of this, I used the DukeGeneralInfoTool. This tool is designed to search through publicly available information from Duke about programs, departments, and other general information. I slightly rephrased your question to "AIPI program degree requirements" when I gave it to the tool to make sure the search was focused on what you wanted to know.

The tool returned ten results which I've now analyzed. These results include links to the program's official page, bulletins describing degree requirements, news articles about the program and student experiences, and information about relevant courses. They're all from official Duke sources, so they should be reliable. I'll use this information to construct a comprehensive answer to your question.

Demo

Main Version

<https://frontend-623186457718.us-central1.run.app>

Web Embedded Version

<https://aipi-590-llm-website.vercel.app/>

Cost Estimations/Approach to Cost Minimization

- Google Cloud Hosting: \$5
- Programmable Search Engine:
 - Look up query: Paid tier: \$5 per 1,000 queries beyond the free tier. \$4.50 (for 1,000 queries per month)
 - Cloud Runs: ~\$0.38 (for 30,000 requests per month):
 - \$0.000024 per vCPU-second.
 - \$0.0000025 per GB-second.
 - \$0.40 per 1 million requests.
 - Conversation Agents or Diag Flow: \$8 (for 5,000 text requests per month)

Shortcomings

- Tools work a little too well, they provide too much information
- Bias regarding the eval, since both a models are gemini based
- Cold start problem.



The background is a dark blue gradient. It features several overlapping circles. In the top left, there is a solid blue circle and a larger circle with a blue-to-purple gradient. In the top right, there is a thin blue outline of a circle. In the bottom left, there is a solid blue circle and a larger circle with a blue-to-purple gradient. In the bottom right, there is a solid blue circle and a larger circle with a blue-to-purple gradient.

THANK YOU!